

Scale-Coupled Dynamics: Gravity and Charge from a Single Non-Commuting Scale Field

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Abstract

We take *scale* — a dimensionless local ratio, not spacetime — as the sole primitive of a formal axiomatic framework, and ask how much of the structure ordinarily assumed of physics can instead be derived rather than posited. Directionality of the scale field, non-commutativity between scalings taken in different orders, and the resulting four-dimensional Lorentzian structure are not built in but forced by the algebra itself, once what counts as a single observation is made precise; curvature, gauge structure, Planck’s constant, and the renormalization group then emerge as different readings of one object rather than independent assumptions joined afterward. The intent is not to unify the forces or to displace spacetime as a working description, but to test how far a minimal, machine-checkable set of axioms can be pushed before it must appeal to structure it cannot itself explain — and to report that boundary honestly, including where the framework currently loses to data or falls silent by its own admission. Every definition, theorem, and proof has been verified end to end in the Lean 4 proof assistant, with zero admitted gaps, so that each claim below can be checked rather than taken on faith.

Keywords: scale invariance, axiomatic physics, philosophy of physics, formal verification, Lean theorem prover, quantum gravity

1 Introduction

General relativity and the Standard Model both take a four-dimensional spacetime manifold as given and place fields upon it. Each has open seams at the place where it meets the other, or meets its own foundations: a singularity where curvature diverges and the classical description simply stops; a cosmological constant whose value is

measured, not explained, by either theory; a renormalization group whose running couplings are computed order by order in perturbation theory rather than obtained in closed form; a particle spectrum whose labels — charge, color, generation — are read off experiment and then built into the group-theoretic structure of the Lagrangian by hand. None of these are failures of either theory on its own terms; they are the places where a choice was made — a manifold, a gauge group, a set of fields — that the theory itself does not derive.

This paper explores a different starting point, one deliberately more austere than either: take the *scale* at a point — a dimensionless ratio of a local unit to a reference — as the only primitive, and ask what follows if that scale is directional, because the physical sources that produce it (energy-momentum) are tensorial rather than scalar, and non-commuting, because performing two scalings along different directions in the wrong order leaves a residue. Spacetime curvature, the gauge connection, particle content, and $\hbar$ are then not independently postulated; they are different questions asked of a single object, the scale vector field, and several of the open seams listed above turn out to be forced closed by that object’s own algebra rather than left as free choices: the dimension and signature of spacetime, the particle-label alphabet, the leading term of the renormalization group, and the relation between the metric connection and the gauge connection all become theorems rather than inputs. Where the derivation closes numerically — the isotropic, gravitational sector alone — it does so exactly and with no free parameter beyond the single dimensionless unit any such theory must carry: a solar light-bending of $1.7515''$ against a measured $1.7509 \pm 0.0002''$, and a Mercury perihelion advance of $42.989''/\text{century}$ against $42.98 \pm 0.04''$ (Section 11 gives the complete confrontation with data, including where the framework currently loses).

This is not a claim that spacetime does not exist, and it is not offered as a competitor to loop quantum gravity, string theory, causal set theory, or any other specific program for going beyond general relativity and the Standard Model. Each of those programs makes a choice about what is fundamental among geometry, matter, and causal structure and then quantizes or discretizes it; each is older, more developed, and reaches questions — a matter content, a mechanism for scattering amplitudes, a route to a working quantum gravity in the loop or string sense — that this framework does not yet reach at all, and Section 13 and Section 14 say so without qualification. What this framework shares with those programs is only the present absence, common to every candidate beyond general relativity and the Standard Model, of a decisive experimental discrimination; naming that shared circumstance is not a claim of parity, only the reason a further candidate, examined to the same standard of rigor and reporting its own failures alongside its successes, is not out of place alongside them. This paper asks a question orthogonal to what any of those programs ask: whether geometry, gauge structure, and particle content are themselves derivable from something more primitive than any of the three — and answers it constructively, by exhibiting the derivation rather than arguing for its possibility in the abstract. The claim is that the four-dimensional, Lorentzian, curved structure usually assumed can instead be *derived* from a single forced algebraic fact: two symmetric operators bracket to an

antisymmetric one. Everything that follows is the working-out of that one fact’s consequences, and nothing else is smuggled in along the way — a claim that Section 13 exists specifically to make checkable rather than asserted.

Every derivation summarized here has an independent, complete proof; we have verified the entire argument mechanically in the Lean 4 proof assistant against no axioms beyond those of standard mathematics (`propext`, `Classical.choice`, and `Quot.sound`), and the Supplemental Material reproduces every statement and proof — 644 theorems in total, spanning definitions, lemmas, and closing numerical bounds — in full mathematical detail, organized to mirror this paper section by section. This paper does not depend on that verification to be read; it is a self-contained physics argument, written so that a reader who never opens the Supplemental Material still receives every derivation’s actual content rather than a pointer to it. What the verification adds is a guarantee that cannot be obtained by proofreading alone: that every “Theorem” in what follows is exactly what it claims to be, with every hypothesis stated and no step skipped, because a proof assistant does not accept an argument it cannot check line by line.

The remainder of the paper is organized as follows. Section 2 states the physical picture in ordinary language before any algebra, because a statement built from commutators and a Lie-algebra classification is not obviously less abstract than the curved geometry it replaces, unless a picture is attached to it first; the picture introduces, informally, every structural result the later sections derive formally, so that the algebra which follows always has a physical question attached to it. Section 3 states the seven axioms precisely. Section 4 derives the forced chain from the axioms to a four-dimensional Lorentzian structure carrying both a metric connection and a gauge connection — the single argument on which the rest of the paper depends. Section 5 states the branch point between the isotropic and anisotropic sectors, which determines whether a given region of the scale field carries curvature alone or curvature together with charge. Section 6 gives the isotropic (gravitational) sector in full: the Schwarzschild solution and the two classical solar-system tests, the Newtonian limit obtained exactly rather than as a truncated series, local covariance under global dissipation, the absence of singularities and horizons, and gravitational-wave dynamics from inspiral through ringdown, including a concrete confrontation with GW250114 [1] and GW241011 [2]. Section 7 gives the anisotropic (particle) sector: the two topological charges, their independence, and the resulting bound on the mass spectrum. Section 8 gives the renormalization group, obtained as a theorem about a defect density rather than a perturbative computation, and the strong-interaction scale. Section 9 gives the quantum sector — $\hbar$ as an exact step along the scale axis — and the all-orders equivalence with general relativity in the gravitational-wave sector that follows from it. Section 10 gives dark matter, dark energy, and the scale-blindness of light. Section 11 confronts the framework with data, in one table. Section 12 states, observation by observation, what would kill which part of the framework. Section 13 is a register of what has been retracted, corrected, cited without proof, and assumed — the standing answer to whether any of the foregoing rests on more than the seven axioms plus one external physical input. Section 14 concludes.

2 The physical picture

Sixteen statements, in the order they compose, with the algebra deferred to the sections that follow. Each previews a result proved formally later; the section reference at the end of a paragraph is where the informal claim becomes a theorem. The picture is stated in ordinary language on purpose, but it should not be mistaken for an easy one: starting from a ratio rather than a place is a genuinely unfamiliar habit of thought, and several of the statements below will read as stipulative before their formal counterparts make them forced. What status to give a framework built this way — whether the axioms should be read as a claim about the world or as the tightest set of commitments that reproduce it, and what it means that a single postulate about observation (A7 below) does real work — is a question this section does not settle; Section 13 and Section 14 return to it once the formal content is on the table.

There is no space, only a ratio. At every point there is a number, and that number is a ratio: the local unit compared to some reference. The choice of reference is unobservable by construction, so only *differences* of this number are meaningful. Picture a map that records, everywhere, how long a meter is there — with no grid drawn on it. The grid, i.e. spacetime, is what emerges from the *inconsistency* between these numbers at nearby points, not something the numbers are written onto. This is a stronger claim than “spacetime is emergent” in the sense usually meant by that phrase: it is not that a smooth manifold approximates some discrete substrate at long wavelengths, but that the manifold structure itself — the fact that nearby points have well-defined relations at all — is read off differences in a single scalar-valued field, with no independent metrical input.

“Two places” is constructed, not primitive. If the scale pattern at two points is parallel, no observable distinguishes them — not two places that look alike, but no second place at all, since every question one could ask of one point has the identical answer at the other. A perfectly uniform region has no internal structure and no extent: distance is the accumulated *non-parallelism* between points, and because the wedge product of two vectors is antisymmetric, separation is naturally a signed area, not a length. Two consequences follow immediately, both proved in Section 4: a truly homogeneous universe would be a single point no matter how large a coordinate volume one assigns it, and the ordinary notion of distance as a positive scalar is recovered only after choosing an orientation, which this framework does not supply for free.

Scale is directional because energy-momentum is tensorial. At each point the scale is not a number but a *vector*: a fluid carries different energy density along its flow than transverse to it, and a tensorial source cannot be matched by a single scalar scale. This is not an aesthetic preference for more structure; Section 6 proves the scalar version of this requirement is not too weak but too strong — it *forbids* the Schwarzschild metric outright, so a scalar scale is not merely insufficient to describe the solar system, it actively excludes the possibility of describing it. Directionality is consequently not an enrichment chosen for generality; it is the minimum needed for the theory to be compatible with a single measured orbit.

Gravity is the ruler changing, not space bending. A change in the *magnitude* of the scale vector across position is exactly gravity; curvature is the second-order inhomogeneity of scale — not a first-order gradient, which is unobservable, but the rate at which that gradient itself changes. Nothing is bent: light follows what looks locally straight, but the ruler used to judge straightness changes along the path, so the trajectory appears curved. Tidal forces are the *second*-order change in the ruler — the first-order, uniform-dilation part is unobservable, being exactly a reference shift. Two bodies in free fall separate or approach not because a force acts between them but because the local unit of length differs infinitesimally at their two positions, and that difference is what a tidal-force meter actually measures; the entire content of the gravitational field, on this reading, is a statement about how a ruler’s length varies from place to place, encoded completely in the second derivative of one scalar field per direction (Section 6). This also settles, rather than merely avoids, the objection that sank Weyl’s 1918 unified theory [3]: there a rescaling one-form was posited independently of the metric, so comparing clocks around two different paths to the same point could give two different answers, contradicting the sharp atomic spectra actually observed. Here the rescaling is not independent — it is the gradient of the one scale field itself — and a gradient’s curl vanishes identically, by nothing more than equality of mixed partial derivatives, so no such path-dependence is possible even in principle, not merely small.

Rotation is direction changing — and it is generated by scaling. A change in the *direction* of the scale vector across position is rotation, and here is the theory’s sharpest forced step: performing two scalings along different directions in the two possible orders must differ by a rotation. This is not a chosen coupling; it is forced by the meaning of the words — a scaling is a pure stretch and hence symmetric, a rotation preserves length and hence is antisymmetric, and the bracket of two symmetric operators is necessarily antisymmetric (Theorem 2). Physically this is the Thomas–Wigner rotation [4], and the Terrell–Penrose effect [5, 6] is its visual form: a fast-moving object is *seen* as rotated rather than contracted, because a change of scale is being read as a rotation. One direction: flat. Two or more: curvature, generated by the scalings themselves, with no separate assumption about how gravity and rotation relate — they are the symmetric and antisymmetric parts of one algebra, and the bracket between them is arithmetic, not physics chosen to fit.

The gauge connection and the gravitational connection are the same object. Both are built from the same log-derivative of the scale pattern, evaluated on two different loci of the same field: where the pattern is directionally uniform (isotropic), the connection reads as curvature; where it is not (anisotropic), the same construction reads as a Yang–Mills field strength. A gauge theory and a metric theory are, in this framework, two questions asked of one connection, not two structures glued together after the fact (Section 4).

“Unifying the forces” is a question in a vocabulary this theory does not have. The standard unification program asks for one gauge group from which every force descends, which presupposes that forces are couplings and that couplings are the fundamental description; in this framework neither holds, since comparisons of local units are primitive and forces are read off them. The question this framework can answer

instead is what is expressible at all: a quantity is observable — invariant under A5's fiducial shift — exactly when it is a function of scale *differences*, not merely typically or approximately, so the scale channel expresses differences and nothing else, and an absolute scale value is consequently not expressible by any construction available here, not merely unmeasured. This is also the reason no single cross-sector test between the scale channel and the rotation channel exists: in the fully scalar reading the two channels are independent by construction, one being an abelianization and the other its kernel, so neither can constrain the other — and Section 4's directional coupling is exactly what breaks that independence once scale is given a direction, replacing "independent channels" with "one channel generating the other."

Whether charge exists at all is itself a fact about scale. The branch between the isotropic and anisotropic loci is not fixed once and for all; it is a property of the scale pattern at a given resolution, and the same region of the substrate can sit on one side of the branch at one energy scale and the other side at a different one. This is the framework's reading of the empirical fact that interactions separate as energy falls: it is not that several independent forces happen to decouple at different scales, but that whether an order parameter, and hence a charge, is present at all is a scale-dependent question with a single underlying source (Section 5).

$\hbar$ is a step along the scale axis. Not a mysterious constant: translating one step along the scale axis fails to commute with position, and the size of the mismatch is $\hbar$ — exactly, not as a first-order approximation in some formal deformation parameter that is later set to a physical value. Uncertainty is then not measurement disturbance but the mutual exclusion of scale sharpness and content richness: a perfectly sharp scale is a single point on the scale axis, and a point carries no content, so resolving any finite amount of structure requires a finite window, and the width of that window times the density of what it must resolve is bounded below (Section 9).

The entire quantum correction to geometry is a rotation, never a stretch. Once the scale algebra is made non-commutative, the correction it produces to curvature is exactly the commutator of two scale generators — and a commutator, by the same symmetric/antisymmetric split that produces gravity and rotation in the classical theory, is purely antisymmetric. The symmetric, radiative part of the geometry is consequently untouched by quantization to any order: gravitational waves are the one observable sector in which this framework and general relativity cannot differ, not merely one in which the difference is currently too small to see. Since the antisymmetric part is exactly the rotational label of Section 7, the same fact says where a quantum-gravitational correction *does* show up first: coupled to spin, not to orbital motion — a specific, if currently unmeasurably small, prediction rather than a generic expectation of smallness (Section 9).

"What exists" is a slice, not a property of the world. Changing resolution does not mean knowing more about the same particles; it means looking at a different set of them. "The list of fundamental particles" is accordingly ill-posed: there is always unresolved content above any scale, with no final layer. The size of the Standard Model's label set measures the logarithmic range it covers, not physical depth — a theory that must remain valid over thirty orders of magnitude in energy needs a label

for everything that appears anywhere in that range, whether or not it is realized at any particular scale one happens to probe.

A particle is an unresolvable knot in the scale vector field. Not a field excitation but a configuration: the vector winds around a point and cannot be combed flat. Its integer charge is the winding number of the vector around a sphere; its mass is not a property it carries but the location at which it appears on the scale axis; its rotational label is $v \wedge \nabla v$, so charge necessarily comes with spin. A second, independent label — a confinement class rather than a winding number — is available from the same construction, and the two are logically independent of each other: knowing one fixes nothing about the other (Section 7). Exactly two labels are available, not three and not one, because the scale pattern in the anisotropic sector has already been shown to collapse to a single vector (Section 4), and a single vector supports exactly a winding number and a monodromy class and nothing further.

The renormalization group is a theorem about counting, not a perturbative expansion. A coupling read as a ratio of defect densities changes with the logarithm of scale at a rate fixed directly by the same dissipation constant that drives cosmic expansion (Section 8). No diagram is summed and no loop order is chosen; the leading term of the beta function is an algebraic consequence of what a coupling *is* in this framework; a subject usually organized around successive orders of a perturbative expansion is organized here around a single exact linear relation.

The universe is not expanding; scale is running. Galaxies are not flying apart — the ruler is shrinking, and because energy scale is the reciprocal of length scale, the energy scale is dissipating. There is no first moment, because there is no zero scale: scale is a unit and is never zero, so “the age of the universe” is not a quantity — what exists is an elapsed ratio, finite and measurable, with no origin (Section 10).

Black holes are arbitrarily dark, never fully black. The horizon is where the time component of the metric vanishes, and that component is minus the square of a scale unit — never zero. No singularity, no horizon, no infinite redshift, all three the same fact: ratios are invertible. A black hole is a region of extreme but finite scale ratio; the exterior geometry is untouched, so shadow, ISCO, and the early ringdown all agree with general relativity, but nothing is causally cut off and no information is destroyed (Section 6).

A horizonless remnant should ring the way a black hole does, then echo. Because nothing in the metric is ever exactly reflecting or exactly absorbing, a gravitational-wave signal that would terminate at a horizon in general relativity instead partially reflects back and forth in this framework, producing a train of echoes evenly spaced in time after the main ringdown, with amplitude falling geometrically from one echo to the next. This is the one place this framework’s predictions are qualitatively distinctive without being quantitatively computable, since the reflectivity itself depends on physics — the interior equation of state — that is explicitly outside this framework’s scope (Section 6).

The two arrows of time are one order. A long-standing puzzle elsewhere is why the direction of entropy increase coincides with the direction of cosmic expansion; the usual answer invokes a special initial condition aligning two otherwise independent facts. Here they are not two facts. Three results, proved independently and never

invoking one another — that time is the direction along which scale drifts (Section 4), that an H -theorem for coarse-grained content holds using only finite resolution, with no scale dependence anywhere in its own proof, and that the entropy of a region of scale ratio R is $\rho \ln R$ (Section 10) — combine into the following.

Theorem 1 (The two arrows are one order) *For $\rho > 0$ and $R_1, R_2 > 0$,*

$$\rho \ln R_1 \leq \rho \ln R_2 \iff R_1 \leq R_2,$$

and likewise with strict inequalities.

Proof $\rho \ln$ is a strictly increasing function of R for fixed $\rho > 0$, since $\ln$ is strictly increasing and multiplication by a positive constant preserves order. $\square$

Ranking regions by entropy and ranking them by scale ratio are *literally* the same order relation, not two arrows that happen to align. The one modeling choice this rests on is reading entropy as $\rho \ln R$, the count of resolvable thresholds within a region of scale ratio R ; that reading is motivated by, but not forced by, the threshold-density construction of Section 8, so the identity is exact given the reading and the reading itself is a choice, stated as such.

Summary. A number (ratio) becomes a vector because its source is tensorial; the vector’s magnitude changing is gravity, its direction changing is rotation; the two are coupled by the meaning of the words involved; the commutator of that coupling gives $\hbar$ and the quantum correction; where the vector cannot be combed flat is a particle; the vector running overall is cosmology. All physics here is a different reading of the same vector field. What this picture does *not* buy is stated equally plainly: no coupling constants, no mass values, no generation structure, no scattering amplitudes. It answers “what,” not “how much”; the quantitative part closes fully only in the gravitational sector, and the particle sector delivers label structure and bounds, not numbers. Whether a picture this abstract — no space, no force, no particle in the usual sense — is owed the interpretation that scale is what actually exists, or only the weaker credit of being the leanest set of assumptions found so far to reproduce what is observed, is taken up directly in Section 14.

3 Axioms

The substrate is a commutative differential ring over n directions; an earlier version of this development posited three distinct carrier structures for different sectors of the theory, later shown to reduce to one. Seven axioms are used throughout.

Axiom 1 (A1 — substrate) *There is a commutative ring A with n commuting derivations $d_1, \dots, d_n : A \rightarrow A$, each additive and satisfying the Leibniz rule.*

Axiom 2 (A2 — scale field) *A distinguished unit $s \in A^\times$, the local scale.*

Axiom 3 (A3 — scale-energy duality) *An energy unit $\varepsilon \in A^\times$ with $\varepsilon \cdot s = 1$.*

Axiom 4 (A4' — directional measurement) *Scale is assigned per direction: $s : \{1, \dots, n\} \rightarrow A^\times$, not a single scalar. (An earlier, scalar version, A4, is refuted in Section 6: it forbids the Schwarzschild metric outright.)*

Axiom 5 (A5 — fiducial covariance) *Physical content is invariant under $s \mapsto u \cdot s$ for any global unit u : only ratios of scale are observable.*

Axiom 6 (A6 — openness / dissipation) *The scale field is not conserved by a global potential: no functional's gradient equals the full drift.*

Axiom 7 (A7 — observability) *A commitment about what counts as a single observation, formalized as a dimension-matching condition between the rotation sector the coupling generates and the direction sector A4' labels; it fixes the number of independent directions rather than leaving it a free integer.*

A7 is deliberately an axiom and not an informal judgment: it is a commitment about what an observation is rather than a claim about the world, and Section 12 returns to the consequence that it cannot be falsified by any single experiment.

One further ingredient, the scale response law

$$e^{2\sigma} R = \kappa \rho, \tag{A6'}$$

relating the curvature scalar R built from the scale field to a matter density ρ , is used in the gravity sector. **It is not one of the seven axioms.** It is an external physical input — the one place general relativity's own field equation is assumed rather than derived — and it is registered as such in Section 13.

This is not a gap the derivation left by oversight; it is the one place where the framework must connect an internal quantity to what is empirically called matter, and every formal theory needs a rule of that kind somewhere. Reichenbach called such a rule a *coordinative definition* [7]: a stipulated link between a theoretical quantity and a physical procedure, not a consequence derived from either side alone. Elsewhere in this framework ρ is already native vocabulary — a threshold density on the scale axis (Section 7) — so A6' is not smuggling in a foreign concept, only fixing how strongly curvature responds to it; that response constant κ is exactly as uncoordinated by the axioms as general relativity's own coupling constant is by its variational principle. The honest claim is not that Einstein's equation has been derived, but that this framework needs exactly one such definition where general relativity needs one, and no more.

One further move inside that single definition is worth naming rather than absorbing silently: elsewhere, content on the scale axis is read positionally — a threshold is a *location*, the answer to “where,” not “how much” (Section 2) — while ρ in A6' is read as a *density*, the answer to “how much is present,” sourcing curvature continuously rather than at isolated locations. A3's duality connects the two in spirit — energy

and scale are already reciprocal — but turning a record of locations into a continuum density that a field equation can source from is a further move this framework does not spell out step by step; A6' takes ρ as already coordinated in its continuum form, not built up from anything else stated in this paper.

4 The forced chain to a four-dimensional Lorentzian gauge structure

This section is the spine of the paper: a single chain of implications, each step forced rather than chosen, from A1–A7 to a four-dimensional Lorentzian structure carrying both a metric and a gauge connection.

The scale/rotation split is not a choice. Given A1–A4', any endomorphism of the tangent data at a point decomposes canonically into a symmetric part (a scaling) and an antisymmetric part (a rotation); this is a fact about the algebra of endomorphisms of an inner-product space, not a modeling decision, since every such endomorphism is the sum of its symmetric and antisymmetric parts and that decomposition is unique. An earlier stage of this development reached the same split by a different route, an abelianization of an abstract comparison group applicable only to the scalar case; that construction is superseded by, and is a special case of, the endomorphism decomposition used from here on, and the two are not independently combined into a stronger joint claim.

Time is the direction along which scale drifts. Given A6 (dissipation), exactly one direction carries a definite, non-cancelling sign of drift; the signature is forced to a codimension-one split, $(1, n - 1)$. A theory in which every direction could drift in either sign with no distinguished one would admit no consistent notion of “before” and “after”; A6 rules that out. Labeling the distinguished direction *timelike* rather than merely *distinguished* is a convention consistent with the split, not a further consequence A6 forces on its own — a distinction registered explicitly rather than smoothed over (Section 13). This is the same direction Section 10 returns to under a different question: given one further reading, that entropy counts coarse-grained content, the direction of drift and the direction of entropy increase turn out to be one order relation rather than two facts pinned together by an initial condition.

The bookkeeping form is $\tau(xy)$ for a trace τ alone, not an independent postulate: any bilinear form on the ring compatible with both the ring product and the derivations must take this shape, because a trace is (up to scale) the unique linear functional invariant under conjugation, and conjugation invariance is what compatibility with the derivations demands.

Curvature is the commutator of transports. Let ∇_i denote parallel transport of a tensorial quantity along direction i , built directly from the scale pattern rather than postulated independently. Then the curvature of the connection is

$$F_{ij} := [\nabla_i, \nabla_j], \tag{1}$$

literally the failure of two transports to commute, rather than an independently defined tensor whose vanishing is imposed or assumed. Two consequences follow

directly from this definition rather than from extra input: first, commuting *derivations* ($d_i d_j = d_j d_i$, part of A1) does not imply a flat *connection* — F_{ij} can be nonzero even when the underlying derivations commute, because curvature lives in the connection, not in the chart; second, the same commutator F_{ij} is what is computed in the gravitational sector as the Riemann curvature and what is computed in the anisotropic sector as the Yang–Mills field strength (Section 7) — **gravity and gauge structure are the same object, read under two different questions about which sector’s scale pattern is being transported.** Neither is assumed; both are the same commutator. This identity extends one step further, to the quantum sector of Section 9: the commutator $[D_i, D_j]$ that gives curvature here is the identical construction that gives $\hbar$ there, evaluated on a different family of transports — phase-space translations in one case, frame transports in the other. Curvature and $\hbar$ are consequently not merely analogous; they are one algebraic construction read on two different transport families, a stronger and more specific claim than any analogy between gravity and quantum mechanics usually offered.

A further split, orthogonal to the isotropic/anisotropic branch of Section 5, separates transport itself into a scale part and a frame-rotation part. The scale part, being a gradient by A5, is automatically integrable — mixed partial derivatives commute — and contributes nothing whatsoever to the curvature F_{ij} ; all of it is carried by the frame-rotation part. This is the structural reason this framework avoids the defect that killed Weyl’s original 1918 unified theory [3], where an analogous non-Riemannian connection produced a length that depended on the path taken between two points (a “second clock effect” ruled out by observation): here the scale transports integrably by construction, so no such path-dependence can arise, while curvature — gravity — lives entirely in the non-commuting frame-rotation sector. Gravity is consequently not, even informally, “the scale field bending something”; it is the failure of frame rotations specifically to commute, with the scale itself never contributing a discrepancy of its own.

Directions are the defining representation, not the algebra itself. Once the algebra generated by the scalings and rotations is named explicitly, its dimension as a Lie algebra ($\dim \mathfrak{so}(3,1) = 6$ for the eventual case $n = 4$) exceeds the number of directions (4), so directions form the vector representation on which the algebra acts, not the algebra’s own coordinates — a distinction an earlier version of this development conflated.

Theorem 2 (Coupling is forced) *Let Σ_i be the scaling generator along direction i . Then $[\Sigma_i, \Sigma_j]$ is nonzero for $i \neq j$ and is purely antisymmetric, i.e. a rotation generator.*

Proof A scaling is symmetric with respect to the pairing induced by the trace form (it preserves direction and only rescales magnitude, so it is self-adjoint); a rotation is antisymmetric (it preserves length, so its generator satisfies $R^T = -R$). For symmetric A, B ,

$$(AB - BA)^T = B^T A^T - A^T B^T = BA - AB = -(AB - BA),$$

so the commutator of two symmetric operators is antisymmetric. That the bracket is additionally *nonzero* for $i \neq j$ — that the coupling is not merely well-defined but genuinely active

— is witnessed concretely rather than argued in general: an explicit boost pair in three spatial dimensions exhibits a nonvanishing bracket, and no general nondegeneracy argument beyond that concrete witness is claimed. $\square$

No coupling constant is chosen here; the coupling exists purely because of what the words “scaling” and “rotation” mean algebraically. The generated rotation is not merely known to exist and be antisymmetric; its concrete form is fixed once the two scaling directions are: for scalings along direction vectors v, w , the rotation their bracket generates is exactly the antisymmetric bilinear pairing $(v \wedge w)_{ij} = v_i w_j - v_j w_i$, the wedge of the two directions. This makes the coupling’s output a genuine two-form — antisymmetric, bilinear, and vanishing exactly when the two scaling directions are parallel — rather than an abstract nonzero element asserted only to exist; a wound configuration in the anisotropic sector, whose direction genuinely varies rather than staying parallel to itself, consequently carries this rotational label automatically, while a homogeneous pattern — one direction wedged with itself — carries none, whatever magnitude it has.

Real rank one, hence the Lorentz family. The Lie algebra generated by one vector’s worth of scalings together with the rotations they force has real rank one: the maximal abelian subalgebra of diagonalizable elements is one-dimensional, since a second independent scaling direction would, by Theorem 2, generate a rotation mixing it with the first, destroying simultaneous diagonalizability. Citing, without reproving, the classification of real-rank-one real simple Lie algebras — $\mathfrak{so}(n, 1)$, $\mathfrak{su}(n, 1)$, $\mathfrak{sp}(n, 1)$, $\mathfrak{f}_{4(-20)}$ — this places the algebra in the Lorentz family $\mathfrak{so}(n, 1)$; the other branches require additional structure (a complex, quaternionic, or octonionic module structure on the directions) absent from a real scale-vector construction, leaving $\mathfrak{so}(n, 1)$ as the surviving case. The step from “one drift functional” to “one-dimensional maximal abelian subalgebra” above is a physical identification of what counts as a single independent scaling direction, not a formal deduction from A1–A7 alone — stated here rather than left implicit, since it is exactly the kind of step where a numerical coincidence can hide a choice (Section 13).

$n = k + 1$, and $k = 3$ from A7. The number of directions n is one (time) plus the number of independent spatial directions k . k is not free once A7’s dimension-matching condition is imposed: requiring the number of rotations the coupling generates to match the number of directions A4’ labels gives $k = 3$. This is not the only matching condition one could write down — comparing the dimension of the full algebra to the dimension of its own defining representation instead gives $k = 2$, and the two conditions provably disagree — so A7 is doing real work: it is a specific commitment about *which* two quantities must be equal, not an arithmetic inevitability. The framework’s own argument for preferring the first condition is that the labels available are the directions A4’ actually assigns scales to, and the coupling’s output must match *those*, not the abstract size of the algebra that generates it — but this is an argument for a modeling choice, not a proof that no other choice is coherent, and it is why A7 is registered as a postulate about what counts as an observation rather than presented as forced by pure counting.

The pattern is one vector; more axes are unavailable. An apparent dilemma between a “uniaxial” and a “biaxial” reading of an earlier theorem is resolved once the

relevant vanishing identity, $v \wedge v = 0$, is recognized as a statement about *homogeneity* — a vector wedged with itself is always zero, regardless of how many axes exist — rather than a constraint on the number of axes. Once that misreading is removed, the scale pattern is shown to collapse to exactly one vector, and additional independent axes are not merely disfavored but structurally unavailable given the coupling of Theorem 2.

The chain therefore delivers, from A1–A7 with no further input: a Lorentzian signature, four dimensions, a metric and a gauge connection that are the same underlying commutator, and a coupling between scaling and rotation forced by algebra rather than fit to data.

5 The branch point

The scale pattern at a point is either *isotropic* — the same in every direction — or *anisotropic*, and whether it is one or the other is a scale-dependent fact, not a fixed property of the point:

isotropic	anisotropic
no order parameter	order parameter present
no charge	charges present
no multiplet	multiplets present
no rotation-sector charge	curvature/rotation active

Both columns have one source: isotropy is a single algebraic condition on the pattern (uniform direction across the pattern’s components), and charge, multiplet structure, and rotation-sector activity all switch on or off together with it, not independently. The formal branch condition itself is a static property of a given pattern, not a function of a separate energy-scale variable; the further, motivating reading — that the same physical region can present as isotropic at one probing resolution and anisotropic at another, giving this framework’s account of interactions separating as energy falls — is an interpretation connecting this branch to the scale-flow structure of Section 8, not itself a theorem proved from the branch condition alone, and is stated here as such. Section 6 treats the isotropic sector; Section 7 the anisotropic one.

An apparent contradiction is worth forestalling: Section 6 computes curvature in a sector called “isotropic.” “Isotropic” here means uniform *direction* of the scale vector, not uniform *magnitude* — a radially varying but directionally uniform scale pattern has no charge and no rotation-sector label, and still has curvature, because curvature is generated by magnitude inhomogeneity (Section 2), which is an axis independent of direction uniformity.

A second consequence of the branch structure fixes what happens *inside* a multiplet, *given* an assignment of one scaling operator to each direction determined by that direction’s own scale value — a hypothesis satisfied automatically whenever such an assignment exists, though real rank one (Section 4) only guarantees one vector and one scaling overall, not a free per-direction assignment, so the statement below is conditional rather than a further automatic consequence of the chain. Under that

hypothesis, two directions assigned to the same multiplet cannot rotate into each other without the underlying scale pattern itself changing first: a scaling generator bracketed with *itself* vanishes identically, and two directions sharing a multiplet by definition share a scale value, hence share a scaling generator, hence generate no internal rotation between them — a binary fact about equal versus unequal scale values, not a count that decreases gradually with multiplet size. Any change in which directions belong to which multiplet accordingly requires a change in the underlying scale pattern itself: multiplet structure cannot rotate on its own, it can only be re-partitioned by the pattern that defines it changing. What is unconditional, established directly in Section 7 without this hypothesis, is that the real-rank-one pattern’s own rotation is carried entirely by its variation, exactly as Section 2 states.

5.1 What “sector” means, and what regresses under which conditions

The word *sector* has been doing two different jobs in this paper, and naming them separately removes what has, in early readings, been the single most confusing feature of this framework. One question is the branch just proved: whether the scale pattern at a point is isotropic or anisotropic. A second, independent question, taken up in Section 9, is whether the underlying algebra is read as commutative (A1’s own baseline) or as its non-commutative, $\hbar$ -deformed extension. Neither question is a different choice of axioms: A1–A7 hold everywhere, unconditionally, and never change. The first question asks about a *value* the axioms’ own objects happen to take at a point — whether a given scale pattern is directionally uniform, a fact about that pattern, not about the rules governing it. The second asks whether one is working in the commutative ring A1 already postulates or in its canonical non-commutative enlargement, a strictly larger structure that contains the commutative ring exactly, as the $\hbar \rightarrow 0$ limit, rather than replacing it. “Sector,” in other words, names a place in a fixed structure, not a different rule set adopted for convenience — which is also why quantum vocabulary appearing inside the gravity section, or charge vocabulary appearing with no $\hbar$ in sight, is not a category error: the two questions are logically independent, and ordinary quantum mechanics, for instance, needs only the second, at any point, isotropic or not. Table 1 lists what each answer, or combination of answers, currently delivers.

The blank last row is worth one further word rather than none: the combination it names — charge-carrying, quantized, genuinely dynamical — is where a reader familiar with the Standard Model would expect quantum electrodynamics or chromodynamics to sit, with particle creation, annihilation, and scattering built from the same $\hbar$ -deformation used in the isotropic sector but applied to an anisotropic pattern instead. Nothing in this paper constructs that combination; the deformation of Section 9 has only been carried through for the isotropic case, and extending it to a pattern that already carries charge is an open problem this framework does not currently solve, not a result being withheld.

That two independent questions asked of one field can look, from outside, like a family of separate theories — mechanics, general relativity, quantum mechanics, a nascent quantum gravity — each with its own domain and its own vocabulary, is not

Condition	Recovers	Status
Isotropic, commutative	Newtonian gravity, then Schwarzschild GR	exact; identical to GR to all post-Newtonian orders in the symmetric part
Any point, non-commutative	Ordinary quantum mechanics (CCR, uncertainty)	exact at first order in $\hbar$
Isotropic, non-commutative	A quantum-gravitational correction to curvature	structural only: purely antisymmetric, coupled to spin, magnitude uncomputed
Anisotropic, commutative	Particle labels, confinement, Yang–Mills-shaped curvature	label structure and bounds, not masses or couplings
Anisotropic, non-commutative	—	not reached: no scattering amplitudes, no quantized gauge theory here

Table 1 The two independent questions “sector” asks, and what each of their combinations currently delivers.

a coincidence this framework is positioned to explain; it is compatible with a broader and older observation in philosophy of science, that physical knowledge often arrives as a patchwork of locally valid models rather than a single seamless law [8]. Nothing here accounts for why that patchwork historically took the shape it did, or resolves any tension between its pieces. What can be said, and no more, is that within this one construction the boundaries between the pieces are answers to two explicit questions about a single field rather than boundaries between different kinds of stuff.

6 The isotropic sector: gravity

This section stays inside a single cell of Table 1 — isotropic, commutative — where the scale pattern points the same way in every direction and the base algebra of A1 is left undeformed. A reader who has just met that table should expect exactly what follows: no charge, no $\hbar$, and a curvature tensor built by hand rather than inherited for free, since the general machinery of Section 4 happens to vanish identically in this cell — the reason is given next — not because different rules apply here than in any other.

6.1 The Schwarzschild solution and the two classical tests

One honest cost of the isotropic sector is worth stating before its results: over a commutative ring the general transport-commutator construction of Section 4 gives *no* curvature at all, since the commutator that defines it vanishes identically on commuting scalars — this is exactly why the isotropic geometry needs its own, separately built curvature tensor rather than inheriting one from the chain’s general machinery, and it is a cost confined entirely to this classical sector, not a limitation on the coupling itself,

the quantum sector, or the anisotropic sector, all of which keep their transports genuinely non-commutative. With the scale pattern isotropic in direction, a metric is built directly from the scale field: writing the local scale along the time and radial directions as s_t, s_r , the line element takes the diagonal form $ds^2 = -s_t^2 dt^2 + s_r^2 dr^2 + r^2 d\Omega^2$. A deformation tensor built directly from second derivatives of the scale field, in the specific combination that the Riemann tensor is forced to equal for a conformally flat metric, gives the Ricci tensor component by component for this static, spherically symmetric ansatz; the vacuum condition (Ricci flat away from a source) becomes the stationarity of a particular sum of scale-derivative terms rather than an independently imposed differential equation. Imposing asymptotic flatness — $s_t, s_r \rightarrow 1$ at spatial infinity, the only boundary condition consistent with A5 (no preferred finite reference scale) — fixes the single remaining integration constant by itself, with no further input: the stationarity condition only pins down the *derivative* of $\sigma_t + \sigma_r$, and asymptotic flatness alone supplies the boundary value that makes the constant zero. A6' plays no role in this particular step; it enters later, in the Newtonian limit and in fixing the overall normalization of the source term. Reciprocity

$$s_t \cdot s_r = 1 \quad (2)$$

follows as a theorem, not an ansatz: the Schwarzschild form is recovered exactly, not approximately, as the unique solution consistent with the axioms and the one external input A6'.

From $s_t s_r = 1$, the PPN parameter is exactly $\gamma = 1$, and the two classical solar-system tests follow with no further assumption:

$$\delta\phi_\odot = 1.7515'' \quad (\text{measured } 1.7509 \pm 0.0002'' \text{ [9]}) \quad (3)$$

$$\Delta\varpi_{\text{Mercury}} = 42.989''/\text{cent.} \quad (\text{measured } 42.98 \pm 0.04'') \quad (4)$$

Both bounds are certified to the precision quoted. Two internally excluded alternatives are worth stating explicitly because they show the framework can fail its own tests: computing deflection or precession from the isotropic sector's magnitude change *alone*, discarding the directional structure of Section 4, gives $0.8758''$ and $14.33''$ respectively — both wrong by a clean integer factor (2 and 3) relative to the measured values. Each is excluded by the same underlying apparatus but through a distinct step: the deflection case is excluded by the theorem that full isotropy together with reciprocity forces flat space outright, while the precession case is excluded by the same $\gamma = 1$ derivation used for deflection, evaluated at the isotropic sector's $\gamma = 0$ instead. Neither exclusion is merely a comparison with data.

6.2 The Newtonian limit and gravitational redshift

Linearizing $s_t = e^\sigma$ about the asymptotic value $\sigma = 0$, the scale response law A6' reduces, to leading order in σ , to a Poisson-type equation for σ sourced by the same density ρ that appears in A6', with the reciprocal constant fixed by the same asymptotic-flatness argument used above. Since $g_{tt} = -s_t^2 = -e^{2\sigma} \approx -(1 + 2\sigma)$, a

weak, static source gives the standard Newtonian potential to leading order, recovering Newtonian gravity as the linear regime of the same reciprocity relation that gives $1.7515''$ at higher order — not a separately posited limit. Gravitational redshift between two static observers at different σ follows directly from the ratio of their local scale units, $s_t(\sigma_2)/s_t(\sigma_1)$, and evaluated for a laboratory height difference this reproduces the Pound–Rebka–Snider result [10, 11] to the precision of Table 2 — an arithmetic check of the A3 reading of scale as a clock rate, not an independent prediction, since it uses the same reciprocity relation already fixed above.

The linear regime above is not an approximation whose error one hopes is small; it is realized *exactly* by extending the substrate ring to the dual numbers $A[\epsilon]/(\epsilon^2)$, a square-zero infinitesimal extension, and setting $\sigma = \epsilon\varphi$ for an ordinary function φ . Every derivation carries over componentwise to this extended ring, and the quadratic invariant $|\nabla\sigma|^2$ vanishes *identically* rather than merely being small, since $\sigma_i^2 = \epsilon^2(\partial_i\varphi)^2 = 0$ by $\epsilon^2 = 0$. Poisson’s equation $2(n-1)\Delta\Phi = \kappa\rho$ therefore holds with no term discarded and no linearization error, for the entirely algebraic reason that there is nothing to discard: the “weak-field limit” is not a truncation of the exact theory but an exact statement about a different, infinitesimally-deformed ring, related to the finite theory only by which numbers one is willing to admit as the value of σ .

6.3 Local covariance survives global dissipation

A6 makes the theory globally non-conservative: there is no functional whose gradient reproduces the total drift of the scale field. Read quickly, this sounds like it should cost something at every point, since a symmetry usually stands or falls as a whole; it does not, because conservation and covariance are claims of different kinds. Covariance is a statement about a single point — that the local physics constructed there has no preferred direction — while A6’s non-conservation is a statement about a sum over every point at once; the second can fail while the first holds everywhere, because nothing about summing infinitely many locally-covariant facts is itself a local fact. General relativity already lives with the same split, in a more familiar guise: local energy-momentum conservation holds identically at every point of an expanding spacetime, $\nabla_\mu T^{\mu\nu} = 0$, while no global, conserved total energy exists for the spacetime as a whole, because no timelike symmetry survives the expansion to define one — a fact learned from cosmology, not a defect of general relativity. This framework’s situation is the same shape: the coupling of Theorem 2 and the Lorentz algebra of Section 4 are pointwise, algebraic facts, holding identically at every point regardless of how the scale field as a whole evolves, so a local Lorentz frame built from the coupling alone is exactly as covariant as in a theory with a globally conserved potential; A6 only denies that any single functional accounts for the drift *everywhere at once*, which a local statement never needed in the first place.

6.4 No singularity, no horizon

Because $g_{tt} = -s_t^2$ with s_t a unit — nonzero everywhere by A2 — g_{tt} never vanishes: there is no horizon and no curvature singularity. The exterior geometry away from the would-be horizon is unchanged from Schwarzschild, so shadow shape, ISCO location,

and the early ringdown spectrum are unaffected; the difference between this framework and general relativity is concentrated entirely in the boundary condition at small radius, where general relativity imposes absorption (a horizon) and this framework cannot, since nothing in it is ever exactly zero.

The absence of a singularity is not merely the absence of a place where s_t hits zero; it is structural at the level of what curvature is built from. The Riemann tensor of Section 4 is a fixed linear function of a deformation tensor built entirely from the *first and second derivatives* of σ — the scale value s itself never appears in the curvature formula, only its rate of change. Two scale fields with identical derivatives at every point therefore have identical curvature even if their values differ arbitrarily, and a region where the scale becomes extreme is on that account alone no more curved than a region where it does not: what general relativity would read as an approach to a singularity, this framework reads as a region where σ is large in magnitude but changing no faster than elsewhere, which is exactly what asymptotic flatness combined with reciprocity $s_t s_r = 1$ guarantees, since a large σ in one direction is bought by an equally large and opposite σ in the reciprocal direction, without either derivative diverging.

A scale ratio, being built from units, is itself a unit — never zero, never infinite, but with no upper bound on its magnitude either. There is consequently no minimum resolvable length and, on the matter side of the same statement, no maximum density (Section 10.5): for any proposed shortest scale a finer one is always available, so there is no final effective description this framework’s own structure ever forces a halt at, only successive replacement at each threshold crossed.

6.5 Two polarizations, from two different sectors, and no breathing mode

A transverse gravitational wave in k spatial dimensions can, a priori, carry perturbations from two independent sources at each point on the propagation plane: a change in the local scale magnitude (the scaling sector) and a change in the local scale direction (the rotation sector), matching the split of Section 4. In the two transverse directions available for $k = 3$, the scaling sector contributes $\dim(\text{transverse}) - 1 = 1$ independent mode — not 2, because a uniform rescaling of both transverse directions together is exactly a fiducial shift, and A5 declares fiducial shifts unobservable, removing that mode from the physical spectrum entirely rather than merely making it small. The rotation sector contributes $\dim \Lambda^2(\text{transverse}) = 1$ independent mode, the wedge of the two transverse directions. The total is

$$1_{\text{scale}} + 1_{\text{rotation}} = 2, \tag{5}$$

matching the two tensor polarizations of gravitational-wave observation, but arrived at as a sum of contributions from two structurally distinct sectors rather than as a single count. This gives a discriminator sharper than the count alone: the framework has **no scalar (breathing) mode at any amplitude**, since that mode is precisely the uniform rescaling A5 removes, whereas scalar–tensor theories of gravity generically predict one at some coupling-dependent amplitude. A breathing-mode detection would

not merely disfavor this framework; A5 gives it nowhere to appear at all, so the correct statement is not “predicted small” but “predicted absent.” This also bears, incidentally, on the ultralight-boson exclusion drawn from GW241011’s rapid spin [2]: that analysis rules out a scalar of Compton wavelength in the superradiant band, and this framework predicts no such field to begin with, since its label structure (threshold, $\mathbb{Z}/2$, $\mathbb{Z}$) has no further slot for one and the scalar polarization it would carry is exactly what A5 removes — the non-detection needs no explanation here, where a scalar-tensor theory owes one.

6.6 Gravitational waves: what radiates, and what does not

Conservation is not an additional postulate laid alongside a field equation for the scale field; it is the condition for such an equation to be writable at all. If curvature is proportional to a source — the shape of any field equation — the Bianchi identity forces that source’s cyclic covariant divergence to vanish identically, with no further input, since Bianchi is itself a consequence of the associativity of transport (the Jacobi identity for the transport bracket). The contrapositive is the sharp form: a source whose cyclic divergence fails to vanish cannot be the curvature source of *any* connection, so this framework does not merely prefer conserved sources over non-conserved ones — it has no expression for a non-conserved source at all. Conservation, read this way from the Bianchi identity as a statement about the scale field’s own algebra, kills the monopole and dipole moments of radiation specifically. This is established purely algebraically: divergences form the additive subgroup generated by images of the non-drift derivations, and because A1’s derivations commute, the drift direction descends consistently to the quotient by that subgroup — no measure, no Stokes’ theorem, and no decay condition at infinity is used; what survives is one purely algebraic condition, that this subgroup of divergences is proper rather than the whole ring. Dipole radiation is accordingly identically zero, consistent with the double-pulsar bound $|\dot{P}_b^{\text{excess}}/\dot{P}_b| < 1.3 \times 10^{-4}$; the quadrupole leads, giving the standard inspiral chirp.

The one constant κ supplied by A6’ appears exactly once, not twice: it fixes both the Newtonian-limit Poisson normalization and the quadrupole radiation coefficient, so fixing it from the static, weak-field limit leaves the radiated power with no remaining freedom — a binary pulsar’s orbital decay is consequently a genuine test rather than a fit to a second free constant. For a circular binary with component masses m_1, m_2 and separation r , the orbital energy $E = -Gm_1m_2/2r$ and the quadrupole radiated power $P = \frac{32}{5}G^4(m_1m_2)^2(m_1+m_2)/(c^5r^5)$ together force the orbit to shrink, $dr/dt < 0$, at a closed-form rate obtained by nothing more than the chain rule applied to $dE/dt = -P$. The gravitational-wave frequency evolution depends on the two masses only through the chirp mass $\mathcal{M} = (m_1m_2)^{3/5}/(m_1+m_2)^{1/5}$, via $df/dt \propto \mathcal{M}^{5/3}f^{11/3}$ — the same combination and the same two exponents, 5/3 and 11/3, that a gravitational-wave detector template actually fits. The chirp mass combination and its two exponents are read off directly from the same quadrupole formula used for the binary-pulsar prediction; the further step from the time-domain separation rate to this frequency-domain form uses Kepler’s third law in its standard, uncontested form, which is cited rather than re-derived here — the exponents are not an independent input chosen to

match observation, but the derivation is not carried through in full from first principles either. Applied to PSR B1913+16 (Hulse–Taylor) [12, 13], the predicted-to-observed ratio of orbital decay is 0.9983, agreeing to two parts in a thousand and bounding any dipole contamination below 0.2%; applied to the double pulsar PSR J0737–3039 [14], sixteen years of timing bound the same dipole fraction below 1.3×10^{-4} , an order of magnitude tighter, with every improvement in timing precision tightening a prediction that has no parameter left to retune.

Theorem 3 (All-order equivalence with general relativity) *The symmetric (radiative) part of the deformation tensor equals its classical, general-relativistic counterpart at every post-Newtonian order.*

This is an exact statement, not a leading-order approximation, and its origin is given in Section 9: the entire quantum correction to curvature is purely antisymmetric, so the symmetric sector carries none of it, at any order. There is accordingly no order at which the two theories’ gravitational-wave predictions could differ — not merely an order too small to currently measure. The gravitational-wave sector can discriminate this framework from general relativity only through the ringdown boundary condition, the presence or absence of a horizon, never through the inspiral or the post-Newtonian expansion itself. This is also why numerical relativity, the full merger, and spin-precession are not computed here: by Theorem 3, an exact treatment of them could only reproduce what general relativity already gives, so building one would be engineering — extending an identity already known to hold — rather than new physical content. What this section computes is deliberately the minimal case that carries the identity’s meaning: leading order, a circular orbit, no spin.

6.7 The horizon-absence signature: echoes and the ergoregion bound

Absence of a horizon permits, but does not predict the amplitude of, late echoes. Writing $A_{\min}$ for the field’s reflectivity at the would-be horizon (a quantity this framework does not compute, Section 11.1) and τ for the light-travel time to that surface, the round-trip delay is $\tau_{\text{echo}} = \tau \ln(1/A_{\min})$ — logarithmic, not linear, in the redshift $1/A_{\min}$: ten orders of magnitude of additional redshift lengthen the delay by only a factor of order ten, not 10^{10} , which is why an echo search looks for a comb of arrivals at millisecond spacing rather than at a Planck time. The n -th echo arrives at $n\tau_{\text{echo}}$, evenly spaced, with amplitude falling as $A_0 R^{2n}$ for reflectivity R after n round trips — a null search excluding an amplitude above some fraction b of the ringdown directly bounds $R^2 \leq b$.

Two independent observational handles exist, and they scale with R differently. Echo searches place a direct bound on R that is polynomial in the excluded amplitude fraction, as above. Ergoregion (superradiant) instability of a rapidly spinning, long-lived compact remnant places an *exponential* bound on $\mathcal{R}$, because a horizonless spinning object whose reflectivity is not exponentially close to unity is unstable on an observationally short timescale — the growth rate of the instability, not merely the

detector’s sensitivity to a faint echo, sets the bound, so a longer-lived remnant tightens the constraint exponentially rather than linearly. GW241011’s [2] primary spin, $\chi_1 = 0.64 > 0.55$ at the reported 90% lower limit, is in the rapid regime, and the remnant’s evident long-term stability makes this the tighter of the two constraints currently available — and, honestly stated, a real and previously unregistered pressure on this framework: because $\mathcal{R}$ is not computable here (Section 11.1), the framework cannot currently say whether it survives this constraint, only that the constraint applies and tightens with every additional rapidly-spinning long-lived remnant observed. The comparison between GW250114’s [1] ringdown and its inspiral-predicted remnant is itself a genuine confirmation, independent of the reflectivity question: the inferred final mass ($62.7 M_\odot$, from components 33.6 and $32.2 M_\odot$ radiating $3.1 M_\odot$, about 5% of the total) and final spin (0.68) predict a ringdown spectrum whose first overtone matches the fundamental-mode inference to within the Kerr-family bound of 30%, and the area law — that the horizon area cannot decrease — is confirmed at 3.4σ across every analysis window and above 5σ in the most sensitive one; both are area-law-family statements this framework’s exterior geometry inherits unchanged from general relativity, since nothing about the exterior solution differs. The damping-time precision of that same ringdown measurement (9%) is looser than the frequency precision (2%) by a factor of about 4.5, which matters because the damping time is exactly the observable a partially reflecting boundary would shift, while the frequency would not — so the ringdown measurement is currently most sensitive in precisely the channel where a reflecting-boundary effect would first appear, and has not yet reached the precision needed to see one.

The magnitude of $\mathcal{R}$ itself requires an equation of state for whatever replaces the interior — a computation explicitly outside this framework’s scope (Section 11.1) — but the qualitative structure, comb-like echoes and an exponential rather than linear bound, is a genuine, falsifiable prediction independent of that missing input.

7 The anisotropic sector: labels and charges

Where the scale pattern is anisotropic, an order parameter, charges, and multiplets appear together, from the same source as the gauge connection of Section 4. A particle is a point defect in the scale vector field — an unresolvable knot — completely characterized by a triple: a threshold (its position on the scale axis, read as mass), a $\mathbb{Z}/2$ confined charge, and a $\mathbb{Z}$ winding charge, and nothing else. This is not an assumption about how many labels a particle may carry; the pattern having collapsed to a single vector (Section 4) leaves two topological invariants available in three spatial dimensions — a winding number (valued in $\mathbb{Z}$, from the vector’s degree as a map to a sphere) and a monodromy class of the order parameter (valued in $\mathbb{Z}/2$, because the order parameter lives in a projective, not a linear, space). That these are the invariants a projective order parameter in three dimensions admits is a standard fact of algebraic topology, cited rather than proved in-repo; what *is* proved directly is the sharper, load-bearing half — that the confinement group is specifically $\mathbb{Z}/2$ and not some other value, which is what rules out the three-valued color identification below.

The $\mathbb{Z}/2$ nature of the confined charge is structural rather than a numerical near-miss: an earlier attempt to read this charge as color required three distinguishable values, and failed against the data by giving two. The order parameter’s home is a projective space, and a projective order parameter simply does not have a three-valued invariant available to it — no adjustment of the construction changes this.

Mass, in this picture, is not a third label alongside the two charges; it is the position on the scale axis at which unresolved content first becomes a distinguishable defect, so “mass” and “threshold” are the same statement read two ways rather than an independent property a particle additionally carries. This is also the precise sense in which there is no final, complete list of fundamental particles (Section 2): resolving further along the scale axis always exposes more threshold structure above whatever has already been resolved, and the claim is not merely that undiscovered particles might exist but that, given an unbounded supply of ever-finer thresholds to resolve — itself a physical premise about the scale axis rather than a theorem derived from A1–A7 alone — the content strictly above any given resolution is never empty, so “the complete particle table” is not a question this framework’s content structure can answer even in principle, for any resolution one names.

7.1 The two charges are independent, and exactly one is confined

The winding charge and the monodromy (block) charge are logically independent: neither is determined by the other, since they are computed from different topological data — a degree and a monodromy class — of the same underlying configuration, and it is possible to exhibit configurations agreeing on one label while differing on the other. This independence is what makes “particle” a genuine pair of quantum numbers rather than one label in disguise, and it is what makes the statement “exactly one of the two is confined” non-trivial rather than automatic: confinement is a further, structural fact about the monodromy charge specifically — a configuration whose block labelling fails to close up after a circuit is not merely rare or energetically disfavored, it is not single-valued and so does not describe a state that can exist in isolation — while the winding charge carries no such obstruction and every winding number is freely realizable on its own. A defect paired with its own opposite always closes up and carries zero net charge in *both* labels simultaneously, independent of which block permutation either one carries, which is the framework’s account of why bound pairs (mesons, in the language of the withdrawn color identification) are always available regardless of the confined partner’s internal labelling.

The same rank-one collapse that limits particles to two labels also bounds the shape of the mass spectrum. If particle masses were drawn from a mixture of two or more structurally distinct populations rather than a single rank-one family, the squared coefficient of variation of the mass distribution satisfies $CV^2 \geq 1$, with equality exactly when the population is the single-family (rank-one) case this framework predicts; a mixture is not merely disfavored numerically, it is excluded whenever the measured CV^2 is below 1. The measured value, 0.875 (Table 2), sits below the bound

and is consistent with rank one, though $0.875 \neq 1$ exactly, so the comparison establishes exclusion of mixtures rather than confirmation of the exact rank-one value — a distinction Section 13 records explicitly rather than eliding.

7.2 Gauge symmetry is the stabilizer of the pattern

Internal symmetry in this framework has no independent space to act on; the only structure at a point is the scale pattern across directions, so any symmetry that is not a motion of the substrate itself must be a relabeling of directions that leaves every local scale value unchanged. The group of such relabelings — the *stabilizer* of the pattern under the permutation action on directions — is not chosen; it is read off the pattern directly, and two directions belong to the same multiplet exactly when they carry the same local scale, so multiplets *are* the degeneracy classes of the pattern and multiplet sizes are the sizes of those classes. Where the pattern is injective (no two directions share a value) the stabilizer is trivial and there is no internal symmetry at all; where the pattern is fully isotropic the stabilizer is the entire permutation group.

This has a direct consequence for what a connection is. A transport between two scale patterns is a relabeling that matches them; where the target pattern is injective, that relabeling is unique — the connection is a function of the scale field alone, with nothing left to choose — and wherever it is not unique, the freedom in choosing it is exactly a coset of the stabilizer, meaning gauge freedom is not a separate structure layered onto the geometry but is *identical to* the degeneracy of the underlying pattern: gauge freedom exists at a point if and only if that point's scale pattern is degenerate there, and its size is exactly the size of the degeneracy. A theory with no coincident scale values anywhere has no gauge freedom anywhere, not as a choice of gauge but as a consequence of the pattern being injective; internal symmetry is consequently never posited as an independent input; it is measured by counting coincidences in a single scalar-valued function per direction.

7.3 Particle lifetimes are scale ratios

A lifetime is a count, not yet a duration: an unstable configuration carries some fixed number of internal ticks before it decays, and that number only becomes a *time* once a tick is given a size. A3 says energy scale and length (hence time) scale are reciprocal, so a tick's size is fixed by whatever energy scale is being used to read it — and an excitation and a laboratory observing it need not agree on that energy scale, exactly as two observers need not agree on the size of a meter before A5 restricts them to comparing ratios. The excitation's own tick is set by its rest energy ε_0 ; the laboratory reads the same excitation with the larger energy scale ε that its motion supplies, so one lab tick corresponds to a smaller fraction of one excitation-tick, and the same fixed count of ticks accordingly occupies *more* lab time, not less. Writing this out, the proper lifetime τ_0 and the lab-frame lifetime τ are related by exactly the scale ratio that appears everywhere else in this theory:

$$\tau = \gamma \tau_0, \quad \gamma = \frac{\varepsilon}{\varepsilon_0}. \quad (6)$$

Time dilation is accordingly not a separate relativistic postulate laid alongside A3, but that same reciprocity read for a moving particle’s own clock rather than for a static observer’s ruler; applied to the CERN muon storage ring measurement [15] this reproduces the measured lifetime to the precision of Table 2.

7.4 Is there a second switch?

The framework predicts exactly one kind of label-opening event: an anisotropic pattern appearing where an isotropic one was, always in the direction of *refinement* — a multiplet only ever splits into smaller pieces as the scale is probed, never the reverse. The framework does not itself prove this refinement direction — only that a change of label structure requires a change of scale pattern — so testing the refinement direction against data is not circular.

This is a different question from Section 8.2’s, and worth keeping separate: that section asks why a formalism spanning a wide range needs many labels at all, and answers that range alone forces a floor on the count regardless of what the labels turn out to be; this one asks whether, within the range already probed, the specific label *types* in use have stayed fixed or multiplied further — a question about this framework’s own inventory, not about range. We count exactly-conserved label *types* (not states) across the directly probed particle spectrum [16], spanning

$$\frac{E_{\max}}{E_{\min}} = \frac{1.3 \times 10^4 \text{ GeV}}{5 \times 10^{-11} \text{ GeV}} > 10^{14}, \quad (7)$$

fourteen orders of magnitude in energy. Over that range the electroweak multiplet only ever shrinks ($2 \rightarrow 1$, a strict refinement) and the color multiplet is unchanged ($3 \rightarrow 3$); no observed symmetry breaking has ever enlarged a multiplet, and no additional exactly-conserved label beyond the four already in the framework’s inventory has been found. This is a genuine, if limited, empirical check: it can only detect *exact* degeneracies opening or closing, and fourteen decades, while large, is finite.

8 Scale flow: renormalization group and the strong scale

A fiducial shift is a symmetry of the geometry (A5) but not automatically of the description of matter: to keep predictions fixed while the reference unit moves, the couplings used to describe matter must move to compensate, and that compensating motion is the renormalization group. Three structural facts follow from A5 alone, before any coupling is specified. First, the scale flow is forced to be a one-parameter group — shifting the fiducial by t and then by u is shifting it once by $t + u$, with no other composition law available. Second, self-similarity forces exponential solutions: any observable obeying the Callan–Symanzik-type equation $dO/dt = \gamma O$ satisfies $O(t) = O(0)e^{\gamma t}$ exactly, the same mathematical fact that produces exact de Sitter expansion in Section 10 — the same equation read at two different scales, not the same numerical constant in both places. Third, the effective-field-theory tower is forced:

each excitation carries a log-threshold, the set of excitations available at a given log-energy is monotone increasing in that energy, and it is exactly constant between thresholds — the description does not merely happen to be piecewise constant, it is provably unchanged wherever no threshold is crossed. These three facts fix the *shape* of any beta function forced by A5: piecewise constant, jumping only where the active content changes. They do not, by themselves, fix the coefficient at any single jump — that remains external input at this level of the argument.

The strong-interaction sector closes that remaining gap for one specific coupling. Reading a coupling not as a diagrammatic sum but as an accumulated *count* — how many resolvable defects a probe at a given log-scale has responded to — A5 forces that count’s density to be constant in log-scale, since a scale-covariant world has no preferred log-scale for defects to bunch around. A constant density ρ integrated over an interval is linear, giving

$$g^{-2}(t) = g^{-2}(t_0) + \kappa\rho(t - t_0), \quad (8)$$

exactly the one-loop running form, with the one-loop coefficient identified as $b = \kappa\rho/2$ — a density of defects per unit log-scale in place of an imported group-theory coefficient (with $\kappa > 0$ retained as a hypothesis, not itself derived). Given that, asymptotic freedom is *equivalent* to $\rho > 0$: $g(t)^2 \rightarrow 0$ as $t \rightarrow \infty$ exactly when there are defects to resolve, so which sign of density gives a free theory is read off the count rather than fitted separately. The combination $\Lambda := \exp(t - g(t)^{-2}/(2b))$ is constant along this flow — a scale-*invariant* axiom system has produced a definite scale not by breaking the invariance in the axioms but by the solution of a scale-covariant equation secreting one, and A5 guarantees no choice of fiducial can shift it. Every hadron mass, read as $c \cdot \Lambda$ for a pure number c , then has a mass *ratio* c_1/c_2 independent of Λ and hence of every free input to the theory, in contrast to the overall scale Λ itself, which is measured rather than predicted, consistent with the one-free-unit statement running through this paper.

One caution is necessary about ρ , and it is the framework’s own internal correction rather than an external criticism. The density entering the running above is *sector-restricted and signed*: a coupling responds only to the structures actually charged under it, not to every threshold in the spectrum, so it is emphatically not the same quantity as the raw threshold density used elsewhere (Section 13) to bound the mass spectrum. A proposed cross-check — measuring this running-coupling density independently from the raw spectral count and comparing — was attempted and found not to be a test at all: the two counts differ by roughly a factor of two (0.864 counting all twelve Standard Model mass scales against 0.444 counting the six quarks alone), and within a single sector the relating equation has one free coefficient and one measurement, so it fixes the coefficient rather than checking anything. This is registered (Section 13) as a proposal that was made and retracted, not as a claim that survives.

Measured against data, the simplest possible ratio law for the confined sector — an equal-ratio geometric tower of defect masses — was checked against lattice glueball ratios [17] (1.39–1.40 and 1.07–1.08 for the two lightest ratios, differing by more than 0.3) and **failed**, refuting that specific hypothesis about the mass law rather than the

parameter-free ratio construction itself (Section 13 records the retraction and why the underlying flow result survives it).

8.1 A second, sharper test: the shape of the threshold spectrum itself

The glueball comparison tests one specific alternative hypothesis using three numbers. A5’s requirement that no log-scale be preferred makes a second, independent, and considerably sharper prediction using the entire spectrum at once. If thresholds are placed by a scale-invariant process — one of constant rate ρ per unit log-scale, with no preferred location for that rate to be higher or lower — the resulting point process has exponentially distributed gaps, and an exponential distribution has squared coefficient of variation exactly $CV^2 = 1$, with no free parameter. The geometric-tower hypothesis predicts the opposite extreme: equal gaps, hence zero variance, hence $CV^2 = 0$ exactly. The two predictions are as far apart as they can be, and this is a statement about the whole gap distribution, not about two or three selected ratios.

Taking the eleven consecutive gaps in $\ln m$ across the twelve distinct Standard Model mass scales, from the electron to the top quark, gives $CV^2 = 0.875$ ($CV = 0.935$) — inside the central 90% sampling range for eleven gaps under the scale-invariant prediction (roughly 0.33 to 1.64 in CV^2), and at the 63rd percentile of it, while the geometric tower’s $CV^2 = 0$ lies far outside that range on the other side entirely. What this establishes, honestly stated, is the sharper half of the comparison: exclusion of the geometric tower is decisive, since 0.875 is nowhere near 0, while agreement with the scale-invariant value 1 is merely encouraging, since eleven gaps is a small sample and the twelve mass scales entering it were selected by hand (massless states excluded, neutrino masses unresolved). The same gap statistics fix the threshold density, $\rho \approx 0.86$ per e-fold — the same ρ_{all} that appears in the retracted cross-sector comparison above, now correctly understood as a property of the full spectrum’s gap statistics rather than a quantity meant to match a sector-restricted running coefficient.

8.2 Label proliferation measures range, not structure

State the formal content precisely before drawing anything from it, because it is easy to overstate. Content resolvable up to log-scale t is the set $\{j : \mu_j \leq t\}$ — the formal counterpart of Section 2’s “no final layer” — and a description is exact between thresholds. Widening the window of validity from t to t' adds exactly the structures whose threshold lies in $(t, t']$ — an identity, since that added set is how “new content” is defined — and two theorems follow with no further assumption: the added content only grows as the window widens (never shrinks, whichever end is moved), and no finite threshold bounds it, since there is always a further structure above any proposed ceiling. Both are statements about a *set* of required labels, not a count, and the honest translation into a count is an inequality, not an equation: a description valid over a range must carry *at least* one label for every threshold that range contains, so its label count is bounded *below* by the range covered. It is not bounded above by it, and cannot be, because nothing here says a chosen labeling scheme is non-redundant — the Standard Model’s own labels (color, generation, weak isospin) are compressed,

human-authored categories layered over many individual thresholded structures, not a one-label-per-structure encoding, so two schemes covering the same range can differ in how many labels they spend on it. What is fixed is only the floor.

That inequality is the formal core of a much older idea in physics, usually left at the level of practice rather than proof: that an effective field theory is not an approximation one is stuck with below some future final theory, but the exactly right amount of structure for the range it is asked to cover, and that no amount of extending it removes the need for new structure above whatever range is currently included [18]. The half of that idea usually taken for granted — that a theory restricted to a scale window loses nothing by the restriction — is the monotonicity theorem read backward: since content is exact between thresholds, a description confined to $(t, t']$ already contains everything resolvable there, and adding a wider range only adds structure the narrower description was never wrong to omit. The half usually argued for informally — that this process cannot terminate in a final, all-scale theory — is the unboundedness theorem stated exactly: for every proposed universal label set there is, provably, a scale above which it is already insufficient. Both halves follow from the same two lines of Lean, not from a general methodological argument about theory-building; that is the sense in which this section formalizes rather than merely restates the case for effective field theory as a permanent feature of physics, not a stopgap.

None of this touches whether the Standard Model’s particular thirty labels are the right ones, or minimal, or free of redundancy between them — only that a fixed, finite label set adequate at every scale cannot exist for any theory built this way, this one included, and that a large label count is accordingly weak evidence for a rich underlying structure and should be read first as evidence about how wide a range the formalism was asked to cover.

9 The quantum sector

Promoting the commutative substrate to a non-commutative one and asking what changes gives the quantum sector. The commutators of the non-commutative scale algebra reproduce the Leibniz rule, the Jacobi identity, the power rule for derivatives, and the canonical commutation relation as theorems rather than postulates. Consequently, A3’s scale-energy duality $\varepsilon \cdot s = 1$ continues to hold exactly only up to a correction of order $\hbar$ in the non-commutative sector — the classical relation is the $\hbar \rightarrow 0$ limit of a relation that is otherwise corrected, not an independent low-energy postulate.

Uncertainty is a general fact about symmetric (self-adjoint) operators on an inner-product space and does not require the scale algebra specifically: for any two symmetric operators A, B and any unit vector ψ ,

$$|\langle \psi, [A, B] \psi \rangle| \leq 2 \|A\psi\| \|B\psi\|, \quad (9)$$

by the Hermiticity of A, B , the triangle inequality, and Cauchy–Schwarz applied to the resulting inner products. Specializing to a commutator that acts as the scalar $i\hbar$ — which is what the canonical Poisson-bracket construction below produces for position and its conjugate momentum, once quantized — gives the Heisenberg relation

$\Delta A \cdot \Delta B \geq \hbar/2$ as a corollary of the general inequality, not as an independently assumed postulate: the constant $\hbar$ enters only through evaluating the commutator, and the inequality itself is pure linear algebra.

Before the exact (Theorem 4) construction, a first-order deformation-quantization [19] version fixes what quantization must mean here. On $A[\hbar]/(\hbar^2)$, a square-zero extension of the substrate ring, a star product is built directly from a biderivation P (a Leibniz bilinear pairing of two derivations, not itself required to be antisymmetric): $(f_0, f_1) \star (g_0, g_1) := (f_0 g_0, f_0 g_1 + f_1 g_0 + P(f_0, g_0))$. This product is associative and reduces exactly — with nothing approximated — to the ordinary commutative product at $\hbar^0$, and its commutator reproduces the *antisymmetric part* of P at $\hbar^1$: $\hbar^{-1}[f, g]_\star = \{f_0, g_0\}_P := P(f_0, g_0) - P(g_0, f_0)$, the Poisson bracket, so two quantities commute in the deformed algebra exactly when this antisymmetric part vanishes on them, and the commutative (gravitational) sector is recovered exactly as the locus where that bracket is zero — P itself need not vanish there, only its antisymmetrization. This is a first-order formal deformation, useful for identifying what non-commutativity must mean before asking whether it can be realized exactly; Theorem 4 below removes the formal parameter and replaces it with an honest additive-and-multiplicative shift map.

Theorem 4 ($\hbar$ as an exact scale step) *There exists a scale shift T — a map on the scale algebra additive and multiplicative, though not required to fix the multiplicative identity — such that T fails to commute with position by exactly one unit: $\Delta X := T(X) - X = 1$, exactly, not as a leading term in an expansion.*

This structure is never itself constructed from A1–A7 alone by the argument above; it asks only for a map with this weaker-than-ring-endomorphism property. We therefore exhibit an explicit model: the polynomial ring $\mathbb{R}[X]$ with d/dX satisfies A1 for $n = 1$, and the translation $X \mapsto X + 1$ is a ring homomorphism realizing T exactly, so the shift is a legitimate, exact, finite — not infinitesimal — translation along the scale axis. This shows the theorem is non-vacuous and the structure compatible with the axioms; it does **not** show that every model of the axioms must carry such a shift, so the identification of $\hbar$ with this step remains an identification, not a derivation, and is registered as such (Section 13).

Theorem 5 (The quantum correction to curvature is purely antisymmetric) *The entire non-commutative correction to the deformation tensor is given by the commutator $[\sigma_i, \sigma_j]$ of scale generators, which is purely antisymmetric.*

This is the origin of Theorem 3: since the symmetric sector carries no quantum correction at all, the gravitational-wave radiative sector is untouched by non-commutativity to any order, while the antisymmetric (rotational/gauge) sector carries the entire quantum correction. This gives a genuine, structural prediction for *where* a quantum-gravitational effect could appear at all, even though its magnitude cannot be: the correction $[\sigma_i, \sigma_j]$ and the rotational label of Section 4 are the same kind of

object, both antisymmetric in the same index pair, so **quantum-gravitational corrections appear first in spin–geometry coupling, not in orbital dynamics** — orbits are computed from the symmetric part, which is exactly the part this correction cannot touch. Stated honestly rather than left implicit: the size of the commutator is set by the same scale step that gives $\hbar$ (Section 9), so relative to the classical terms in the solar system this correction is utterly negligible, and nothing here predicts an effect within reach of current instruments — what is claimed is the structure of the leading correction and precisely which channel it can and cannot appear in, not a measurable number.

10 Dark matter, dark energy, and light

10.1 Dark matter: two routes, making opposite predictions

This framework offers not one account of dark matter but two, and it does not choose between them — an honest state of affairs rather than an oversight, since the two are logically independent and observationally distinguishable.

The first is a detection-threshold mechanism, requiring no branch-point argument at all. A detector is itself a physical system sitting at some energy scale ε_D ; by A3 that scale is the reciprocal of its spatial resolution, so what it can register is not “an interaction” but an interaction that moves its own state by at least its threshold θ . An excitation whose own energy scale ε sits far below ε_D deposits a fraction $\varepsilon/\varepsilon_D$ of the detector’s scale — strictly positive, so the interaction genuinely happens, but driven below threshold whenever ε is small enough, so the observation genuinely fails. Gravity, meanwhile, couples to the total energy $N\varepsilon$ of a population of such excitations, which is blind to how that total is partitioned among them. A large multiplicity of very-low-scale excitations is consequently gravitationally loud and non-gravitationally silent *simultaneously and for the same population*, for any fixed gravitational signal and any fixed detector: the theorem constructs, for any target mass M and any detector threshold, a scale and multiplicity reproducing M gravitationally while sitting strictly below that threshold. On this route dark matter has thresholds and hence a genuine mass spectrum, and a sufficiently well-matched probe would eventually see a signal.

The second is the isotropic-trajectory mechanism of Section 5: a region whose scale pattern is isotropic in direction still has curvature, since magnitude inhomogeneity, not direction inhomogeneity, sources gravity, so such a region gravitates with no anisotropic order parameter and hence no electromagnetic or confined charge (Section 7) at any scale whatsoever — not a small coupling waiting to be resolved, but no label to couple to in the first place. On this route dark matter has no thresholds and hence no mass spectrum and no species to enumerate: it is one structural condition on the scale pattern, not a family of candidate particles with different masses and couplings, and it forbids exactly what the first route permits — a charge-type self-interaction, since that requires the directional order parameter this route’s matter provably lacks.

The two routes make **opposite** predictions and the framework states this as a fork rather than resolving it: the low-scale route gives a coupling that is small but strictly

nonzero at every finite separation, hence a spectrum in principle detectable by a sufficiently sensitive or well-matched probe; the isotropic route gives a rotational label that is identically zero at *every* scale, hence collisionless behavior and no spectrum to find by any amount of improved sensitivity. A null result at ever-improving direct-detection sensitivity favors the isotropic route; a spectral feature or a self-interaction signature favors the low-scale route; the framework hosts both constructions and does not adjudicate between them, which is a weaker but more honest position than committing to either.

10.2 The direction of time from finite resolution alone

The entropy ordering used in Theorem 1 rests on an H -theorem that uses nothing about scale in its own proof, which is precisely what makes the later identification with the scale-drift arrow a substantive fact rather than a definition chasing its own tail. The construction is elementary: let X be a finite set of microscopic configurations and $\pi : X \rightarrow Y$ a resolution map, generally non-injective because observation happens at finite resolution. The *saturation* of a subset $S \subseteq X$ under π is the set of all configurations π cannot distinguish from something in S ; a configuration set is always contained in its own saturation, so coarse-graining can only add configurations, never remove them, giving $|S| \leq |\text{sat}(\pi, S)|$. Microscopic evolution, being a bijection (the substrate is reversible — nothing in A1–A7 privileges a direction of time at the microscopic level), preserves the cardinality of any subset exactly. Composing one step of reversible evolution with one step of coarse-graining therefore cannot decrease the number of configurations compatible with what is observed: entropy, read as the log of that count, is non-decreasing by counting alone, with no statistical assumption beyond finite resolution — and once it has strictly increased at one step, no entropy-non-decreasing law can undo the evolution, which is the precise sense in which a reversible microscopic law produces an irreversible macroscopic one, matching the status of the Boltzmann equation.

This also gives the framework’s structural reading of Sakharov’s third condition for generating any asymmetry at all: departure from thermal equilibrium. In a closed system that departure is a fine-tuned episode that must be arranged; under A6 the universe never stops dissipating, so the condition is satisfied permanently rather than accidentally — strict entropy production is itself a departure from equilibrium, since every observable is stationary in equilibrium by definition, so an open, dissipating universe satisfies Sakharov’s condition at every moment it produces entropy at all. This is deliberately modest: it establishes that the necessary condition holds throughout this framework’s history, not that any particular observed asymmetry (baryon or otherwise) follows from it.

10.3 Dark energy is the dissipation rate, squared

Cosmic expansion is, in this framework, not the motion of anything through space but the running down of the total dimensionless energy scale ε , read through the duality A3 as a growing length scale. Dissipation at a constant fractional rate λ has a unique solution, $\varepsilon(t) = \varepsilon_0 e^{-\lambda t}$ — no ansatz is assumed, the exponential is forced by

the differential equation alone — and the corresponding scale factor $a = 1/\varepsilon$ is exact de Sitter expansion, $a(t) = a_0 e^{\lambda t}$, with constant Hubble rate $H = \lambda$ and $\ddot{a} = \lambda^2 a > 0$: the expansion accelerates for any nonzero dissipation rate, with no repulsive substance posited anywhere. “Dark energy” is accordingly not a substance with negative pressure; it is this dissipation rate, and matching the observed density gives $\Lambda \propto \lambda^2$ (specifically $\rho_\Lambda = 3\lambda^2/8\pi G$ once λ is identified with the asymptotic Hubble rate) — the *square* of the dissipation rate, since Λ enters through the acceleration $\ddot{a}$ rather than through λ itself, not the dissipation rate directly.

Pure exponential dissipation is the sharpest and most falsifiable case, not the only one available: generalizing $\dot{\varepsilon} = -\lambda\varepsilon$ to a power law $\dot{\varepsilon} = -\lambda\varepsilon^q$ for a single dimensionless exponent q fixes the equation of state exactly, $w = (2q - 5)/3$, invertibly, $q = (3w + 5)/2$. The case $q = 1$ is exactly $w = -1$ with no freedom at all — pure exponential dissipation — and it is the only value of q that gives $w = -1$, not one case among several that happen to agree. This is the framework’s sharpest exposure in this sector: if $w = -1$ is excluded by observation, pure exponential dissipation is dead, and the surviving theory must have $q \neq 1$, read off the data by the inversion formula. What the power-law generalization does *not* provide is a mechanism for w to *evolve*: for any fixed q , w is still a constant, since q is a single dimensionless exponent rather than a function of time, so DESI’s [20] preference for an evolving w_0, w_a remains a tension with the shape of the construction itself and not merely with the value $q = 1$ — a constant-but-shifted w is available to this framework in principle, a genuinely time-varying one is not, which is why the DESI tension of Section 11 is correctly read as a tension with A5’s constancy requirement rather than with a single fittable number.

10.4 Light is scale-blind, but redshift is not

Reciprocity $s_t \cdot s_r = 1$ (Section 6) holds pointwise, not only in vacuum, so the null cone defined by $g_{tt}g_{rr} = -1$ is invariant under any global rescaling $s \mapsto u \cdot s$ permitted by A5. Since a photon’s trajectory is defined purely by the null condition, its propagation is insensitive to the overall scale — the light cone is scale-blind by the same fiducial covariance that makes any single scale unobservable. This gives photon dispersion identically zero at every energy, a stronger and structurally distinct statement than merely “dispersion smaller than currently measurable,” and it is the basis for the Fermi-LAT comparison of Table 2 [21].

This appears to contradict the cosmological picture directly above: if light cannot see the scale field at all, how can a redshift measurement ever register cosmic dissipation, which is exactly a change in that field? And Section 6’s local-covariance result sharpens the puzzle rather than resolving it, since it proves a spatially constant cosmic drift changes *nothing* measured locally. Both results are theorems, so both are true, and the appearance of contradiction is a sign that two different things were being called by the same name. They were. A null vector has no length, but assigning it an *energy* is not a property of the vector alone — it is the pairing of the vector with an observer’s own direction, and an observer is made of matter, which (unlike light) does carry the scale. The *causal structure* light defines is scale-blind, exactly as above; the *energy* an observer assigns to a light ray is not, because assigning it requires a scale-carrying observer in the first place. Redshift is consequently not a property of light at

all: it is the *ratio of two observers' scales*, $(u_1/u_2)^2$, with light contributing nothing to that ratio beyond carrying the comparison from one event to another. This simultaneously resolves the apparent tension with local covariance: a redshift measurement is not a local measurement of the drift, it compares the scale at emission with the scale at absorption, and that is a scale *difference*, which A5 declares physical, while local covariance concerns what a single observer measures at one point — cosmology observes exactly the quantity the axioms permit and no more.

10.5 Questions only this framework can ask

Everything so far has answered questions posed elsewhere: does the theory give Schwarzschild, does it give a graviton-like sector, does it give an area law. That is the right first test, and every section above has run it. It is not the right last one, because a framework built on different primitives should be able to *ask* things the usual vocabulary cannot pose at all, and whether it can is where it either earns its keep or is exposed as a paraphrase of existing physics in new notation. Seven such questions follow from the primitives already in use — a scale that is a unit, a pattern that collapses to one vector, a single global drift, the antisymmetric wedge, a threshold density, the isotropic/anisotropic dichotomy, and one commutator — each pushed for what it demands to know rather than for what it was built to answer.

Separation is not primitive; non-parallelism is. With no space as a primitive, “two places” cannot be assumed — it must be constructed, and the wedge is the only construction available (Section 2). The question this opens, unavailable in a theory that starts from a manifold: what is the observable that measures *degree* of separation, if not a distance? The wedge answers it, and being antisymmetric, the framework’s own separation is necessarily *oriented* — a signed area, not a length, which is not in the standard vocabulary at all.

There is one switch, not one per interaction. The isotropic locus has no rotational label, no charge, no multiplet, and no curvature simultaneously, because all four have one source (Section 5). The framework’s own question: the usual account has separate energy scales at which separate interactions become distinct; here there is one boundary and then refinement. So — is there any label whose appearance is *not* a refinement of an already-anisotropic pattern? A genuinely independent switching-on would falsify the single-source claim, and it is a question no other framework needs to ask, because no other framework has only one switch to defend (Section 7.4 runs this test against fourteen decades of data).

Content belongs to intervals, never to a single scale. A window of zero width resolves nothing; with threshold density ρ , a window of width Δ contains $\rho\Delta$ structures, vanishing with the window. “What exists at energy E ” is consequently malformed as a question in this framework: what exists is a property of a scale *interval*, never of a point. The complementarity this produces — sharpness in scale opposed to richness in content — is not Heisenberg transplanted; the conjugate pair is scale-sharpness against content-count, and the smallest window in which a single structure is unambiguous is exactly $1/\rho$.

There is no global field equation, and A6 is why. A field equation can be written only if its source obeys a cyclic conservation law, and A6 says the universe’s global

source is not conserved — so there is no field equation for the universe as a whole, only local ones, which survive because local covariance survives global dissipation (Section 6). The framework’s own question: if nothing global is being solved, what fixes the large-scale history at all? The only candidate is A5 itself, a symmetry rather than a dynamical law — the large-scale history is fixed by what is *unobservable*, not by what evolves, and whether that is sufficient is an open question this framework poses that general relativity, built around a global field equation, cannot.

The quantum-gravity experiment is an ordering test, not a particle search. The entire quantum correction to geometry is a commutator of scale gradients (Section 9), and a commutator is an order discrepancy: the native experiment is not “detect a graviton” but “compare the scale along two directions, in both orders, and look for a discrepancy.” The magnitude of that discrepancy is set by the scale step and sits far below anything currently measurable, stated here rather than hidden; what is distinctive is that the observable this framework proposes is an *ordering*, not an amplitude, and no particle needs to exist for the question to be well posed.

Defects change parity in pairs, with no coupling constant. The confined charge’s $\mathbb{Z}/2$ label composes multiplicatively, so the total parity of any collection cannot change under binding or splitting. This is not a selection rule imposed on a Lagrangian — there is no Lagrangian here — it is arithmetic in $\mathbb{Z}/2$: is there any process that changes total parity? The framework says no, categorically, which is a cleaner statement than a selection rule derived from an assumed symmetry, since there is no symmetry being assumed and hence nothing to break.

There is no maximum density. A ratio of scale units is itself a unit — never zero, never infinite, but unbounded in magnitude — and matter, in the anisotropic sector, *is* anisotropy of the scale, so there is no maximum matter density, no Planck-scale density cutoff, and no scale at which the description must be replaced by a qualitatively different kind of theory. This is the same statement as the absence of a minimum length (Section 6), seen from the matter side rather than the geometry side, and it is the framework’s sharpest structural disagreement with programs that posit a fundamental cutoff.

None of these seven is a translation of a question asked elsewhere; each follows from a primitive the standard vocabulary lacks. Their honest status varies and is stated here rather than left implicit: the one-switch question and the parity question are testable now in principle; the interval-content complementarity is already tested through the resonance-width bound of Section 8; the no-maximum-density question is testable only through its contrapositive (a confirmed minimum length would falsify it); the separation, global-history, and ordering-test questions are conceptual claims with no current measurement attached. None of the seven is a prediction of a number.

11 Confrontation with data

Table 2 is the single scoreboard of this paper.

One live tension and one live open question are not fatal — both sectors are under active measurement — and the last four rows are options the framework itself rejects; these are more informative than the passing rows because they demonstrate

Quantity	This framework	Measured	Status
<i>Derived from the axiom chain (plus $A6'$); certified to the precision quoted</i>			
Solar light deflection	$1.7515''$	$1.7509 \pm 0.0002''$	agrees
Mercury perihelion advance	$42.989''$	$42.98 \pm 0.04''$	agrees
PPN γ	1 (exact)	$ \gamma - 1 \lesssim 2 \times 10^{-5}$ [22]	agrees
<i>Arithmetic check of the $A3$ reading, not an independent prediction</i>			
Muon time dilation	$64.43 \mu\text{s}$	$64.378 \pm 0.026 \mu\text{s}$	agrees (10^{-3})
Pound–Rebka	2.455×10^{-15}	$(2.46 \pm 0.03) \times 10^{-15}$	agrees
<i>Converted; λ itself is measured — the one free unit</i>			
Dark energy density	$5.845 \times 10^{-27} \text{ kg/m}^3$	$5.86 \times 10^{-27} \text{ kg/m}^3$ [23]	agrees
<i>Currently live tests</i>			
w non-evolution (forced by $A5$)	$w = -1$, no running	DESI: $w_0 > -1$, $w_a < 0$	tension , 2.8–4.2 σ
$f_0(500)$ pole Γ/m	< 1.157	CCL 1.234 [24] / GKPYP 1.090 [25]	open
Photon energy-dependent delay	exactly zero	$E_{\text{QG}} \gtrsim 7.6 E_{\text{Pl}}$	passes, distinct prediction
Gravitational wave speed	exactly c , no dispersion	$ \Delta v/v \lesssim 10^{-15}$	passes
GW polarization count	exactly 2, no breathing mode	network polarization tests	passes
Dipole radiation	identically 0	double pulsar $< 1.3 \times 10^{-4}$	passes
Ringdown echoes	reflectivity $\neq 0$	GW250114 residual = noise	open, under pressure
Ergoregion instability	reflectivity bound uncomputed	GW241011 $\chi_1 = 0.64$, long-lived	open, tighter pressure
Ultralight scalar	no such slot	$10^{-13}\text{--}3 \times 10^{-12} \text{ eV}$ excluded	compatible
Area-law analogue	scale ratio never decreases	consistent with observation	passes (not discriminating)
Mass spectrum CV^2	$= 1$ (rank one; mixtures need ≥ 1)	0.875	partial
Exactly-conserved unconfined charge	exactly one	electric charge, $B-L(?)$	undecided, awaits $0\nu\beta\beta$
Black hole entropy	$\propto \ln R$	—	unmeasured (area law also unmeasured)
<i>Excluded by the framework itself — the most informative rows</i>			
Equal-ratio mass tower	$\text{CV} = 0$	0.935	excluded
Isotropic-sector-only deflection	$0.8758''$	$1.7509''$	excluded (factor 2)
Isotropic-sector-only precession	$14.33''$	$42.98''$	excluded (factor 3)
Color = block charge	$\mathbb{Z}/2$ (two values)	needs three values	excluded (structurally)

Table 2 The complete numerical scoreboard.

the framework is capable of being wrong internally. On timescale: even a seesaw-based lepton-number-violating model predicts a $0\nu\beta\beta$ half-life far above current exclusion limits [26], so this is a genuine but slow-fused test.

11.1 Equation of state is out of scope

The magnitude of the ringdown reflectivity $\mathcal{R}$ (Section 6.7) requires an equation of state for the interior structure replacing a horizon, which varies enormously between candidate compact objects and which this framework does not attempt to compute — this is explicitly outside its scope, not a gap it intends to close. What the framework *does* deliver without an equation of state is the qualitative echo/ergoregion structure of Section 6.7 and the comparison between the linear and exponential strength of the two observational bounds.

12 How to falsify this framework

Observation	Kills	Status
Kills the whole framework		
w evolves with redshift (5σ , dataset-consistent)	A5, dimensionless-ness, all geometry	DESI $2.8\text{--}4.2\sigma$, but BAO alone self-consistent
Energy-dependent photon delay, any order	A3 + light-cone scale-blindness	$E_{\text{QG}} \gtrsim 7.6E_{\text{Pl}}$, winning
GW dipole radiation	the conservation theorem, hence Bianchi	double pulsar $< 1.3 \times 10^{-4}$
GW breathing mode	A5 (a breathing mode is a fiducial shift)	tensor favored over scalar/vector/mixed

Table 3 No adjustable parameter absorbs any of these four: a detection is fatal, not a call for retuning.

A7 cannot be falsified by this table. It is a commitment about what counts as an observation; anyone may reject it and obtain a different k , and no experiment adjudicates that choice — which is precisely why it was promoted to an axiom rather than left an informal judgment. The one free unit is likewise not falsifiable: it is the scale in which predictions are expressed, not itself a prediction.

What does *not* constitute falsification: the absence of a Standard Model particle table (Section 7 argues this was never a goal); the absence of computed hadron masses (the same free unit again); and, emphatically, agreement with general relativity in the gravitational-wave sector — Theorem 3 proves the symmetric sector agrees at every order, so no discrepancy is expected there in principle.

Observation	Kills	Status
Kills one sector		
A minimum length exists	horizon, singularity and max-density results	same bound as photon dispersion
$f_0(500)$ pole $\Gamma/m > 1.157$ confirmed	Section 7’s resolvability bound	CCL above, GKPY below
$B-L$ gauged and exactly conserved	the “exactly one charge” result	$0\nu\beta\beta$ not seen, slow fuse
A label opens that is not a refinement	Section 7.4’s single switch	no counterexample in 14 decades
Total-parity-violating process	the $\mathbb{Z}/2$ arithmetic	none known

Observation	Exposes	Status
Forces an admission of what cannot be computed		
Rapid spinning compact object decays via ergoregion instability	horizon-absence requires small reflectivity, uncomputed	GW241011 $\chi_1 = 0.64$, long-lived; tightest current pressure
Ringdown echoes seen	same, reversed: predicted to exist, amplitude not predicted	null searches so far
BH entropy $\propto$ area, not $\ln R$	the log-entropy result	entropy itself unmeasured

13 Register: retracted, corrected, cited, and assumed

A theory should carry its own history of correction, not erase it. This section is a standing answer to the question “does this rest on the axioms alone?”

Retracted (4). A geometric mass-tower law, later shown to be an unproved assumption rather than a derived consequence, and refuted by lattice glueball data under that assumption; a color identification that required three values where the projective order parameter structurally provides only two; a proposed cross-sector consistency test comparing two quantities that were not, on inspection, the same quantity; a codimension analysis that described a circle-valued rather than directional scale.

Corrected in scope (6). In each case the underlying result survives and the physical interpretation attached to it does not — most notably, an earlier reading in which the rotational and scaling sectors were treated as independent channels is true only of the scalar version of the theory; the directional scale makes the rotation sector *generated by* the scaling sector (Theorem 2).

A recurring error (5 occurrences). Five of the corrections above share one root cause: scalar reasoning applied where the structure is directional. The diagnostic

that catches it: whenever a claim mentions a count — one axis, one scale, one direction — ask a count of *what*; value and direction are never the same count.

Cited, not proved (2). The classification of real-rank-one real simple Lie algebras invoked in Section 4, and the relevant homotopy facts about projective spaces used in the particle-charge construction of Section 7. If either citation fails, the weaker statement proved before invoking it still stands; only the identification built on top does not.

Assumed, registered rather than smuggled (3). A6', the scale response law, external to the seven axioms; a uniform-density assumption needed for one dimension-counting step in Section 4; A7 itself, promoted to an axiom rather than left implicit.

Structures now witnessed (6 of 6). Six structures carrying load-bearing theorems — most importantly the scale-shift structure carrying Theorem 4 — were, as found by external audit, never constructed from A1–A7 anywhere in the original development, making the associated theorems true but conditional. All six are now witnessed by explicit models: the polynomial-ring construction of Section 9 witnesses the scale-shift structure itself and a cosmic-history example; the remaining four are witnessed by separate, unrelated concrete models — a matrix trace, an adjoint action on a general ring, a real-valued translation flow, and a constant function — chosen for whichever structure each was needed for, not by one uniform construction. This shows the structures are *compatible* with the axioms, not that the axioms *force* them — the gap between compatible and forced is real and is not closed by exhibiting a witness.

What rests on the axioms alone. The scale/rotation split, the codimension-one signature split, the bookkeeping form, the coupling (Theorem 2), real rank one (subject to the physical identification just above), conservation from Bianchi, the full gravity chain through 1.7515'' and 42.989''/century, the label structure of the anisotropic sector, every statement in Section 10.5 including the one-arrow-of-time identity (Theorem 1). $\hbar$ as an exact scale step is deliberately absent from this list: it is witnessed, not forced, exactly as stated above.

14 Conclusion

Taking scale rather than spacetime as primitive, a single chain runs from axioms to numbers. Scale must be directional, since a scalar version forbids the Schwarzschild metric outright. Once directional, the scale–rotation coupling is forced by the meaning of the two words: scaling is symmetric, rotation is antisymmetric, and the bracket of two symmetric operators is necessarily antisymmetric. The coupling, together with the drift being a single functional, gives real rank one, hence the Lorentz family, hence a fixed dimension with three spatial directions set by A7 — a commitment about what counts as an observation, not a free parameter. The metric connection and the gauge connection turn out to be the same commutator, read under two different questions. The isotropic/anisotropic branch determines whether labels exist at all, itself a scale-dependent fact, giving this framework's account of why interactions separate as energy falls. Curvature is scale deformation, the renormalization group is the same drift read as a running coupling, $\hbar$ is a step along the scale axis, and the entire quantum correction to geometry is purely antisymmetric. Dark matter is the isotropic trajectory of

the same field that produces ordinary gravity, and dark energy is that field’s own dissipation rate. The gravity sector closes quantitatively at $1.7515''$ and $42.989''/\text{century}$ with nothing assumed in between.

These are not metaphors; every claim stated as a theorem in this paper has an independent, complete proof, verified end to end by machine — 644 checked theorems, no admitted gaps, and no hidden axioms beyond those of standard mathematics. But the statement that this framework reproduces known physics cannot currently be made without qualification: the scoreboard carries one live tension, DESI’s $2.8\text{--}4.2\sigma$ preference against w non-evolution, which is a tension with A5 itself and not with a fittable parameter, since no version of this framework admits an evolving w ; and one live open question, the side of the resolvability bound on which the true $f_0(500)$ pole lies. Nor does the framework explain the Standard Model — Section 7 argues that was never the goal — and Section 13 records four retracted claims, six claims corrected in scope, two citations of results not reproved here, and three assumptions registered rather than hidden, along with one recurring scalar-for-directional error caught five times.

What a fully verified argument can establish is exactly one thing: that these conclusions follow from these axioms, with no gap in between. Whether nature adopts the axioms is a separate question that this paper does not, and should not be read to, answer — which is exactly why the scoreboard and the register are given in full.

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- Data availability: Not applicable — this work is a formal mathematical development; no experimental data were generated.
- Materials availability: Not applicable.
- Code availability: The complete Lean 4 formalization (644 machine-checked theorems, zero admitted gaps) is available in the project repository.
- Author contribution: X.Y. conceived the theory, wrote the formal Lean 4 development, and wrote the manuscript.

Appendix A Note on the Supplemental Material

The Supplemental Material gives, in the same order as the sections above, a complete mathematical restatement of every one of the 644 machine-verified theorems underlying this paper — full formal statements and full proofs, organized to mirror the structure of this paper section by section, with each item individually identifiable for cross-reference. This paper does not depend on the Supplemental Material to be read;

the Supplemental Material exists so that every claim made informally above can be checked in complete formal detail.

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